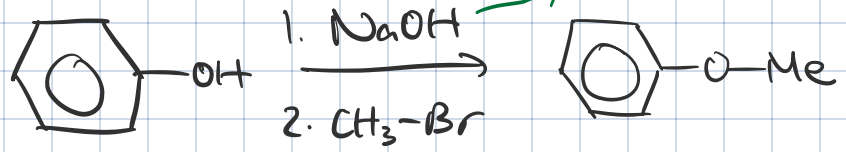
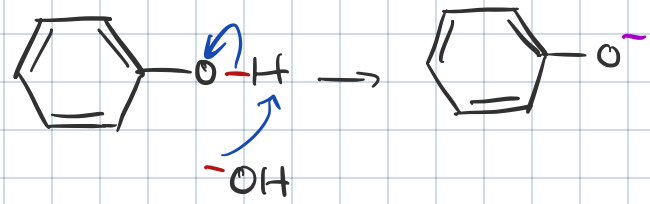
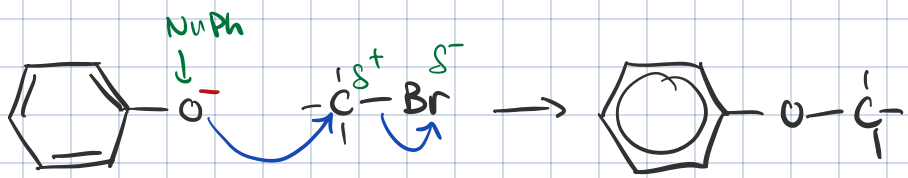
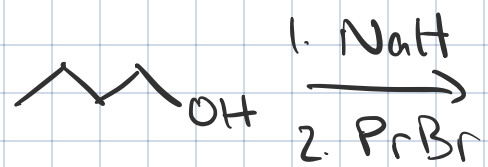
Williamson Ether Synthesis

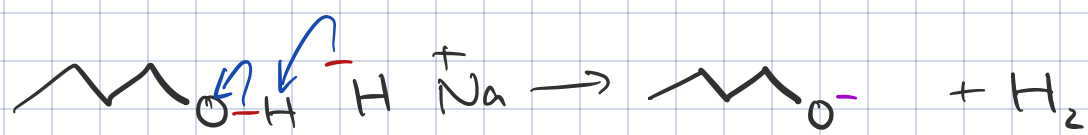


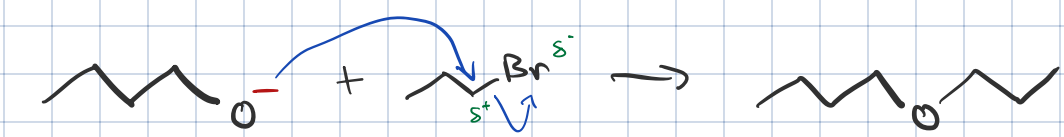




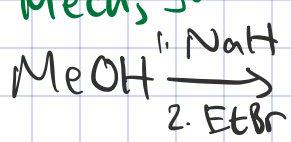
1. Alcohol deprot. → Alkoxide
2. Alkoxide is strong base, SN2 on the alkylhalide
   * If alkoxide is bulky, may be E2
   * 1° alkylhalide will favor SN2, otherwise, there will be competing rxn's
   * can be intra-molecular

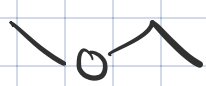




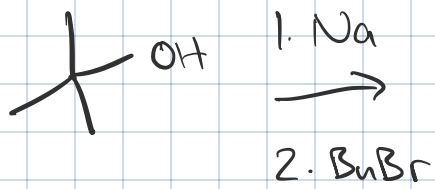


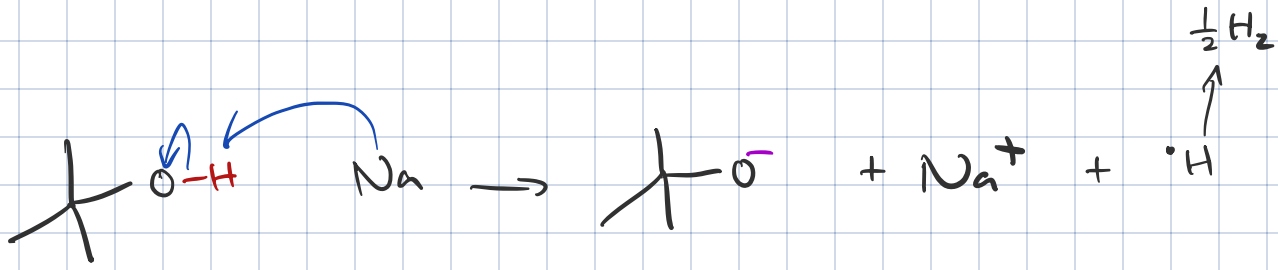
No Mechs, just predict

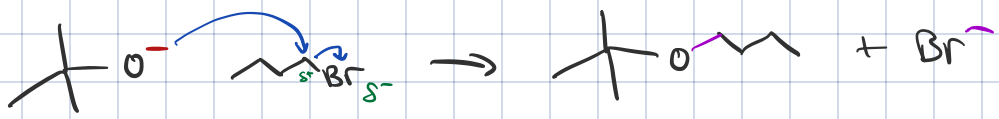


1. $MeO^- + H_2$
2. MeOEt ⟶ 

Mech

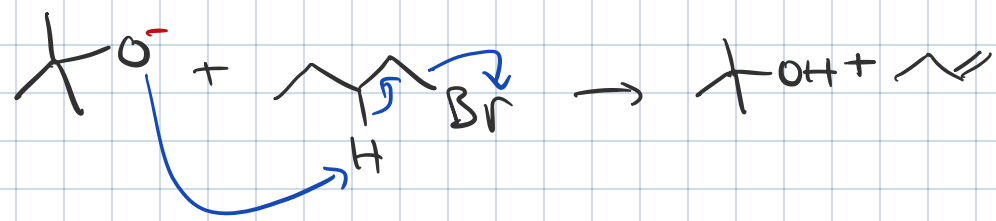


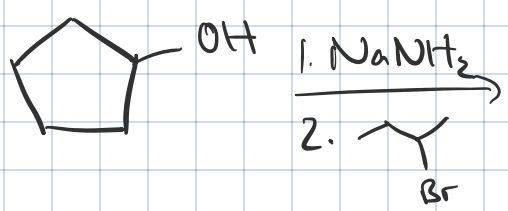


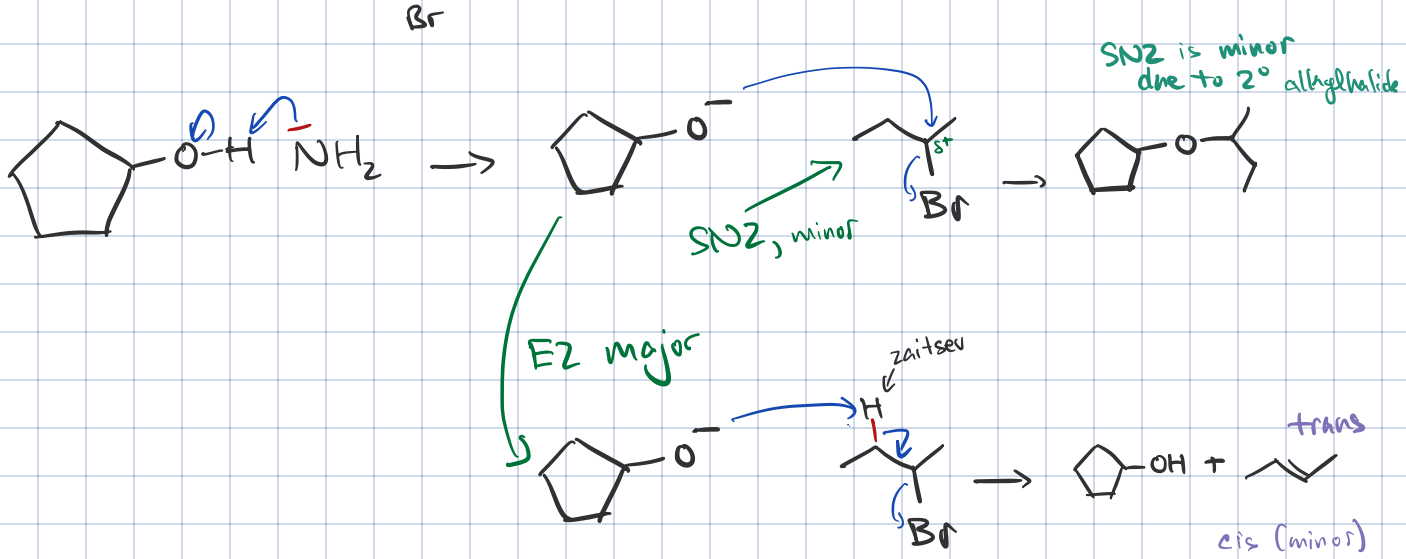


↳ or (b/c bulky base)

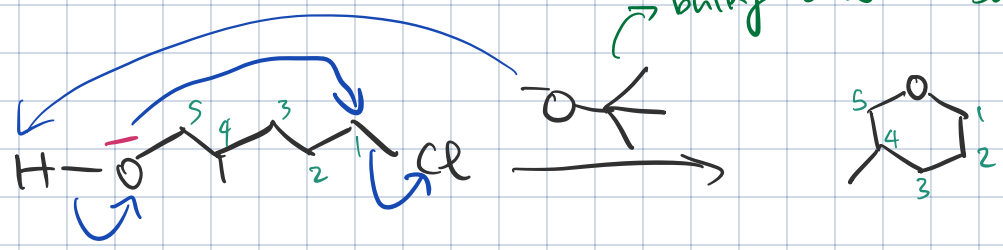
↳ E2 rxn







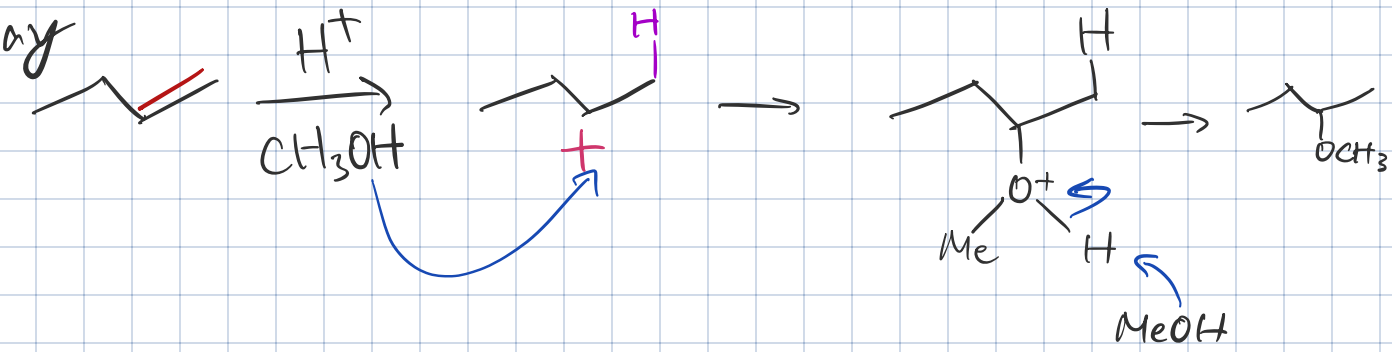
bulky base = bad NuPh



check stereochemistry

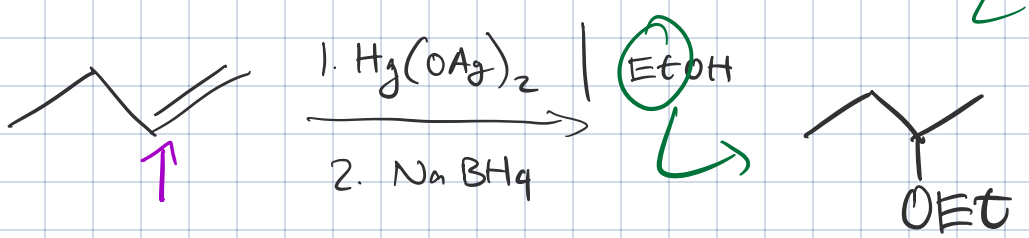
Making Ether from alkene

1st way



2nd way alkoxy mercuration demercuration

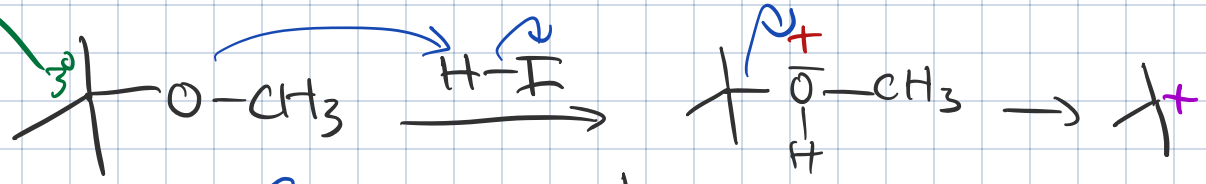
Markiunikov RXN



Ethers don't react in Base, but in acid, they have acid-catalyzed cleavage

SN1

Step 1:



Step 2:

If HI is an excess, CH_3I is formed by protonating MeOH & SN2 w/ I^-

↳ 1°